



> build six things that see the invisible

Six EE Projects That See The Invisible

GNSS, direction finding, magnetic anomaly, and secure comms — six builds from the sensing and signals side of defense. From a receiver that hunts a hidden transmitter across a field to a link that scatters across the band and encrypts itself. All receive-only or protective; real defense-grade signal processing.

REVISED EDITION — now with build renders, updated bills of materials, and wiring for every unit

01	GNSS Receiver & Resilience Pull position and time out of signals from 20,000 km up — and learn how they're attacked	\$50
02	RF Direction Finder Point a needle straight at a hidden transmitter — the core of signals intelligence	\$185
03	Magnetic Anomaly Detector Sense a hidden vehicle by the dent it makes in Earth's magnetic field	\$52
04	Seismic Unattended Ground Sensor Feel footsteps and vehicles through the ground before you can see them	\$42
05	Drone Detection Receiver Spot a drone by the radio it can't stop transmitting — passive counter-UAS	\$500
06	Frequency-Hopping Encrypted Link A radio link that scatters across the band and encrypts every packet — resilient comms by design	\$46

● By Enes · @1nska · Electrical Engineer — a Build List field guide

◆ PRESENTED BY BLUEPRINT

From a hardware idea to a full build plan — in one prompt.

Every unit in this field guide was designed, sourced and rendered in **Blueprint**, the AI hardware design tool. Describe the build in plain English and get wiring diagrams, an itemised BOM with live sourcing links, and a step-by-step assembly guide with a 3D render — the six units here started exactly this way.

▸ start your build, free — blueprint.am

GNSS Receiver & Resilience

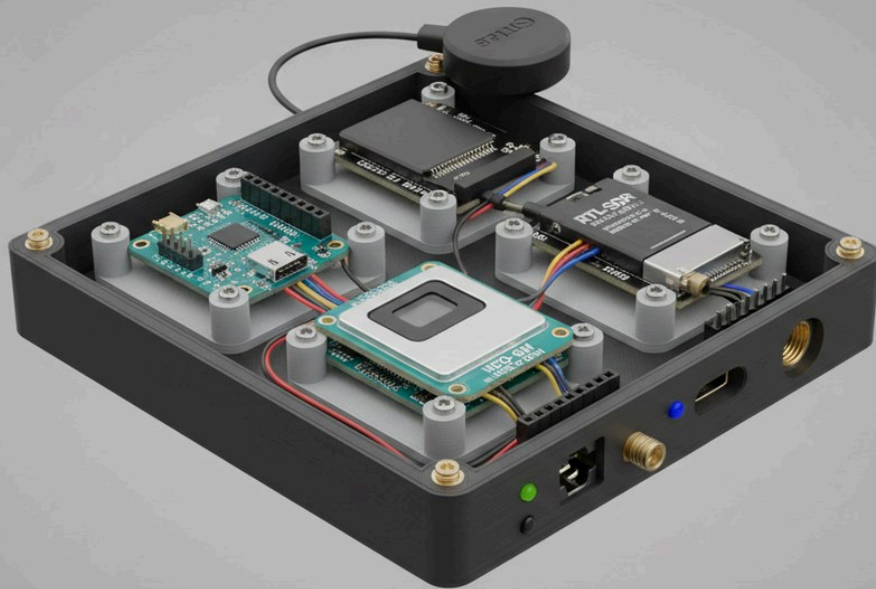
Pull position and time out of signals from 20,000 km up — and learn how they're attacked

\$50

beginner

1 weekend

↗ open in Blueprint



POWERED BY **Blueprint**

As-built desktop unit — ESP32 + u-blox NEO-6M + RTL-SDR Blog V4, GPS patch antenna on the lid.

cost — ~\$50 as built • the NEO-6M core alone is ~\$20

why this matters

Every ship, aircraft, drone, and vehicle depends on GNSS for position and precise time, which makes it one of the most important — and most contested — signals in defense. Building a receiver is the way in: a \$12 module locks onto satellites 20,000 km overhead and hands you latitude, longitude, altitude, and nanosecond-grade time, and with an SDR you can even glimpse the raw signal buried below the noise floor. Understanding how GNSS works, and why it is so fragile, is foundational to positioning, navigation, and timing (PNT) engineering — one of the largest and most strategically vital areas of defense EE.

WHAT THIS PROVES TO A RECRUITER

You understand GNSS fundamentals (trilateration from time-of-flight, the role of precise time), NMEA data, why the signal sits below the noise floor and how correlation recovers it, and the concepts of jamming and spoofing resilience. PNT is core defense and aerospace EE.

→ the job this maps to: positioning, navigation, and timing (PNT) engineering — aerospace, defense, autonomous-systems, and timing-infrastructure teams.

! RECEIVE ONLY — NEVER JAM OR SPOOF

This build only listens. Jamming or spoofing GNSS is a serious federal crime and endangers aircraft, ships, and emergency services, so nothing here transmits and the resilience discussion stays conceptual. Understanding the threat is a defensive skill; creating it is not.

how it works

- ▶ **Trilateration from time:** Each satellite broadcasts its position and an extremely precise time. The receiver measures how long each signal took to arrive, converts that to a distance, and solves for its own position where the ranges intersect — needing at least four satellites to also solve for its own clock error.
- ▶ **Time is the real product:** Because position comes from timing, GNSS is first a time system. The satellites carry atomic clocks and the receiver recovers time to nanoseconds — which is why GNSS quietly synchronises power grids, cell networks, and financial systems, and why time itself is a defense priority.
- ▶ **Below the noise floor:** The signal arriving at your antenna is weaker than the background noise. The receiver recovers it by correlating against each satellite's known pseudo-random code (a spread-spectrum trick) — a huge processing gain that pulls the signal up out of the noise.
- ▶ **Reading the output:** A receiver module outputs standard NMEA sentences over serial: position, altitude, speed, satellites in view, and fix quality. Parsing these is your window into what the receiver is actually doing.
- ▶ **Why it is fragile:** The received power is tiny, so a small interferer can jam it, and because the format is public, a fake stronger signal can spoof it. Real systems defend with directional and adaptive antennas, multi-constellation receivers, and spoofing detection — the resilience side of PNT.

bill of materials

COMPONENT	PART / SPEC	QTY	~COST
ELECTRICAL			
Main Controller	ESP32-WROOM-32 Development Board	1	\$8.00
GNSS Receiver	u-blox NEO-6M GPS Module (GY-NEO6MV2)	1	\$12.00
Software Defined Radio	RTL-SDR Blog V4 USB Dongle	1	\$25.00
PRINTED PARTS & HARDWARE			
Enclosure Base	3D-printed part	1	\$1.50
Enclosure Lid	3D-printed part	1	\$0.50
MCU Sled	3D-printed part	1	\$0.20
GNSS Module Sled	3D-printed part	1	\$0.15
M2.5 Board Screws	M2.5x6mm Machine Screws	8	\$0.16
M3 Enclosure Screws	M3x10mm Socket Head Cap Screws	4	\$0.12
M3 Threaded Inserts	M3x4mmx5mm Brass Heat-Set Inserts	4	\$0.20
Software Defined Radio Mount	3D-printed part	1	\$2.00
Build total (as specified)			\$49.83

♦ BOM, wiring & render generated in **Blueprint** — component links via [blueprint.am](#)

wiring at a glance

SIGNAL & POWER CONNECTIONS

- ESP32 UART2 ↔ NEO-6M — MCU TX2 to GNSS RX, GNSS TX to MCU RX2
- NEO-6M power: 3.3 V + GND from the ESP32 (~100 mA)
- Active GPS antenna → NEO-6M, SMA pass-through in the lid
- RTL-SDR Blog V4 → host over USB (optional raw-L1 viewing)

build steps

1 Get a fix

Wire the GNSS module to your microcontroller over serial, take it near a window or outside, and wait for a lock. Watch the satellite count climb and the first position fix appear.

2 Parse NMEA

Read the NMEA sentences and extract latitude, longitude, altitude, speed, satellites in view, and fix quality. Print them cleanly and confirm they match your real location.

3 Log a track

Record position over time and plot it (a walk around the block). See the accuracy, the noise, and how it degrades indoors or under cover.

4 Recover precise time

Use the receiver's timing output (the PPS pin, if present) and note how tightly it disciplines time — the feature that makes GNSS a synchronisation backbone.

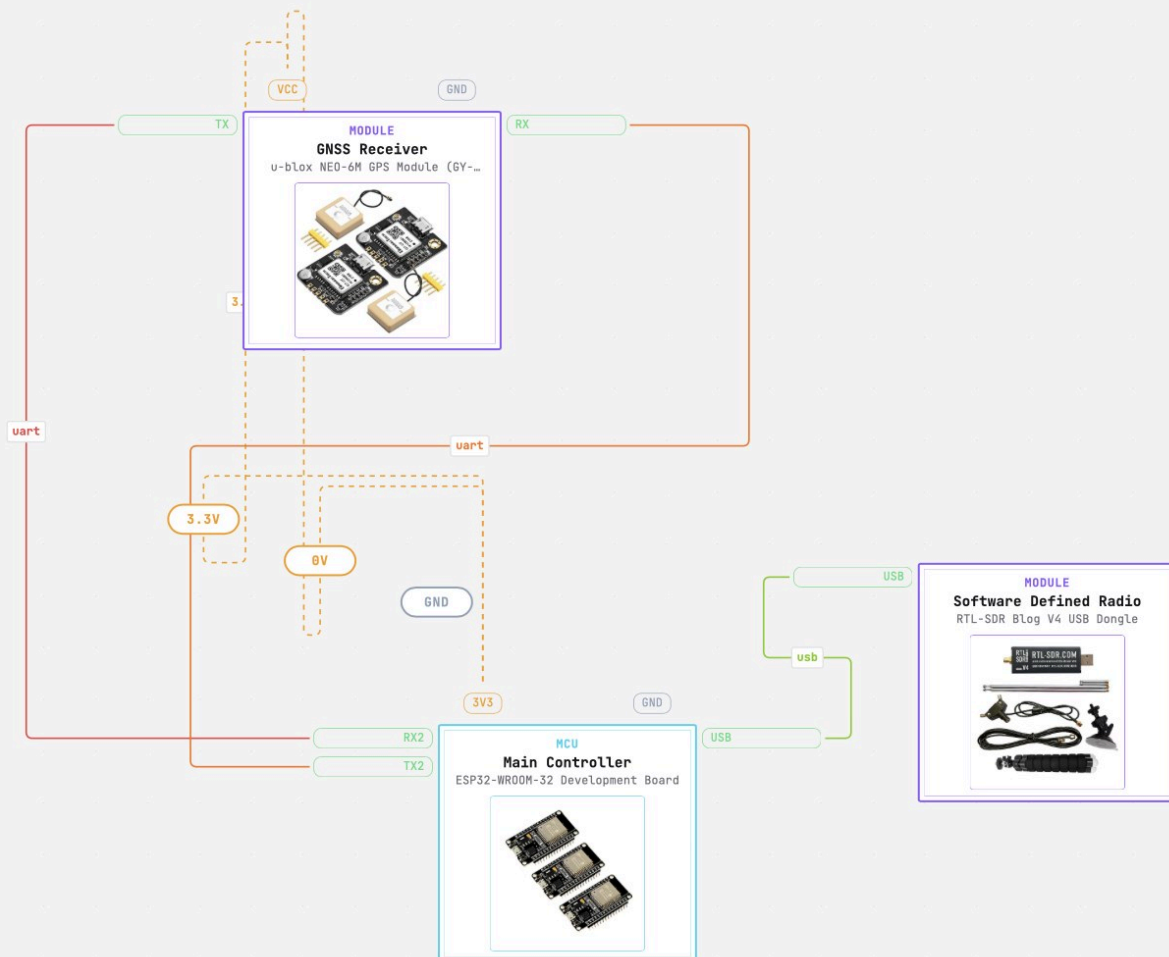
5 See the raw signal (optional)

With an RTL-SDR and an active antenna, tune to L1 (1575.42 MHz) and observe what looks like pure noise — then read how correlation pulls the real signal out. That gap between 'looks like noise' and 'carries the world's timing' is the whole lesson.

interview questions

- How does a GNSS receiver determine its position from satellite signals?
- How is the GNSS signal recovered when it sits below the noise floor?
- Why does a GNSS fix require at least four satellites, not three?
- What is the difference between jamming and spoofing, conceptually?
- Why is GNSS fundamentally a timing system?
- How do modern receivers improve resilience against interference?

full wiring schematic



signal & power graph — generated in [Blueprint](#) · [open this build in Blueprint](#) ↗

RF Direction Finder

Point a needle straight at a hidden transmitter — the core of signals intelligence

\$185

advanced

1-2 weekends

➔ open in Blueprint



POWERED BY **Blueprint**

As-built rotational DF station — 144 MHz Yagi on a tripod, RTL-SDR Blog V4 feeding a Raspberry Pi 4.

cost — ~\$185 as built (Pi 4 + 144 MHz Yagi + tripod) • from ~\$45 with an RTL-SDR, Yagi and your own laptop

why this matters

Finding where a radio signal comes from is one of the oldest and most important jobs in defense electronics — search and rescue, spectrum enforcement, electronic warfare, and signals intelligence all rest on it. Build a direction finder and you can watch a bearing swing to point straight at a hidden transmitter across a field. The accessible version uses a single SDR with a directional antenna and signal strength, or a pseudo-Doppler array; the serious version uses a coherent multi-channel SDR to compute true angle-of-arrival from phase. It is the marquee build of this guide — visual, deeply technical, and a direct window into how SIGINT and EW systems locate emitters.

WHAT THIS PROVES TO A RECRUITER

You understand direction-finding techniques (RSSI / rotational, pseudo-Doppler, and phase-based angle-of-arrival), coherent multi-channel reception, antenna arrays, and the math that turns antenna phase differences into a bearing. Coherent RF and DF are advanced, in-demand skills across SIGINT, EW, and spectrum work.

➔ the job this maps to: signals intelligence, electronic warfare, and spectrum engineering — defense RF, spectrum-monitoring, and search-and-rescue teams.

! LOCATE, DON'T INTRUDE

This is receive-only: you locate transmitters, you never transmit or interfere. Use it for fox-hunting, tracking down interference, or search-and-rescue beacons, and respect privacy and radio law — don't use it to covertly track people.

how it works

- ▶ **The basic idea:** A directional antenna hears a signal loudest when pointed at its source. Rotate it (or steer a phased array electronically) and the direction of peak strength, or of a specific phase relationship, points back toward the transmitter.
- ▶ **RSSI and rotational DF:** The simplest method reads received signal strength as you rotate a Yagi; the peak is the bearing. Cheap and intuitive, but reflections and multipath in real environments blur it, which is why better methods exist.
- ▶ **Pseudo-Doppler:** Rapidly switching between antennas arranged in a circle imposes a Doppler-like modulation whose phase encodes the arrival angle. It works with a single receiver and is a classic, affordable DF technique.
- ▶ **Phase-based angle-of-arrival:** With multiple coherent receivers sharing one clock, you measure the tiny phase difference of the same signal arriving at spaced antennas. That phase difference maps directly to the arrival angle — the method behind precise, modern DF.
- ▶ **From one bearing to a fix:** A single bearing is a line; take bearings from two or more locations and where they cross is the transmitter's position. Moving receivers, or multiple stations, turn a direction into a location.

BUILD NOTE

Start cheap: one RTL-SDR and a directional (Yagi) antenna plus signal strength, or a pseudo-Doppler array. A coherent KrakenSDR is the upgrade that gives true phase-based angle-of-arrival — far more robust, and worth it once the basics click.

key design decisions

- **Why coherent multi-channel over a spinning antenna?** A rotating Yagi with RSSI is cheap but slow and easily fooled by reflections, which create false peaks in real environments. Coherent receivers measure phase across a fixed array instantly and are far more robust to multipath — which is why professional DF uses phase-based angle-of-arrival, not a spinning antenna.
- **Why does coherence (a shared clock) matter so much?** Phase-based DF depends on comparing the exact phase of the same signal at several antennas. If the receivers don't share a clock, their phases drift independently and the comparison is meaningless. Locking them to one reference is the entire reason a coherent SDR works where independent dongles do not.
- **Why does one direction finder only give a line, not a point?** A single station measures the angle a signal arrives from, which defines a bearing line — the transmitter is somewhere along it. Only by crossing bearings from different locations (or moving) does the intersection reveal the actual position, which is why DF networks and moving platforms exist.

bill of materials

COMPONENT	PART / SPEC	QTY	~COST
ELECTRICAL			
SDR Receiver Module	RTL-SDR Blog V4	1	\$30.00
VHF Directional Antenna	144MHz 3-Element Yagi Antenna	1	\$60.00
Raspberry Pi 4	Raspberry Pi 4 Model B 4GB	1	\$55.00
USB Power Bank	Anker PowerCore 10000	1	\$20.00
PRINTED PARTS & HARDWARE			
Stationary Tripod	Amazon Basics 50-inch Lightweight Tripod	1	\$18.00
Yagi Tripod Adapter	3D-printed part	1	\$1.50
SDR Protective Case	3D-printed part	1	\$0.75
1/4-20 Mounting Screws	1/4-20 x 1/2 in. Machine Screws	2	\$0.70
Cable Management Zip Ties	4-inch Nylon Cable Ties	10	\$0.50
Build total (as specified)			\$186.45

♦ BOM, wiring & render generated in **Blueprint** — component links via [blueprint.am](#)

wiring at a glance

SIGNAL & POWER CONNECTIONS

- VHF Yagi (144 MHz) → RTL-SDR RF input via coax (SMA), shield to GND
- RTL-SDR BLog V4 → Raspberry Pi 4 over USB (data + 5 V)
- Anker 10,000 mAh bank → Pi 5 V / GND
- GNU Radio / DF software on the Pi logs RSSI vs. heading

build steps

1 Pick a method and a target

Choose RSSI-with-Yagi (cheapest), pseudo-Doppler, or coherent DF. Pick a legal, known test transmitter — a friend with a handheld, or a licensed beacon — to hunt.

2 Build the antenna or array

Build or buy a Yagi for your band, or assemble the circular antenna array for coherent DF. Antenna geometry directly determines your accuracy.

3 Get a bearing

For RSSI, rotate and log signal strength versus heading; the peak is the bearing. For coherent, run the DF software and read the computed angle-of-arrival. Confirm it points sensibly at the known source.

4 Handle multipath

Move around and watch reflections create false bearings, then average, change location, or switch to a phase-based method to firm up the estimate. This is where DF gets real.

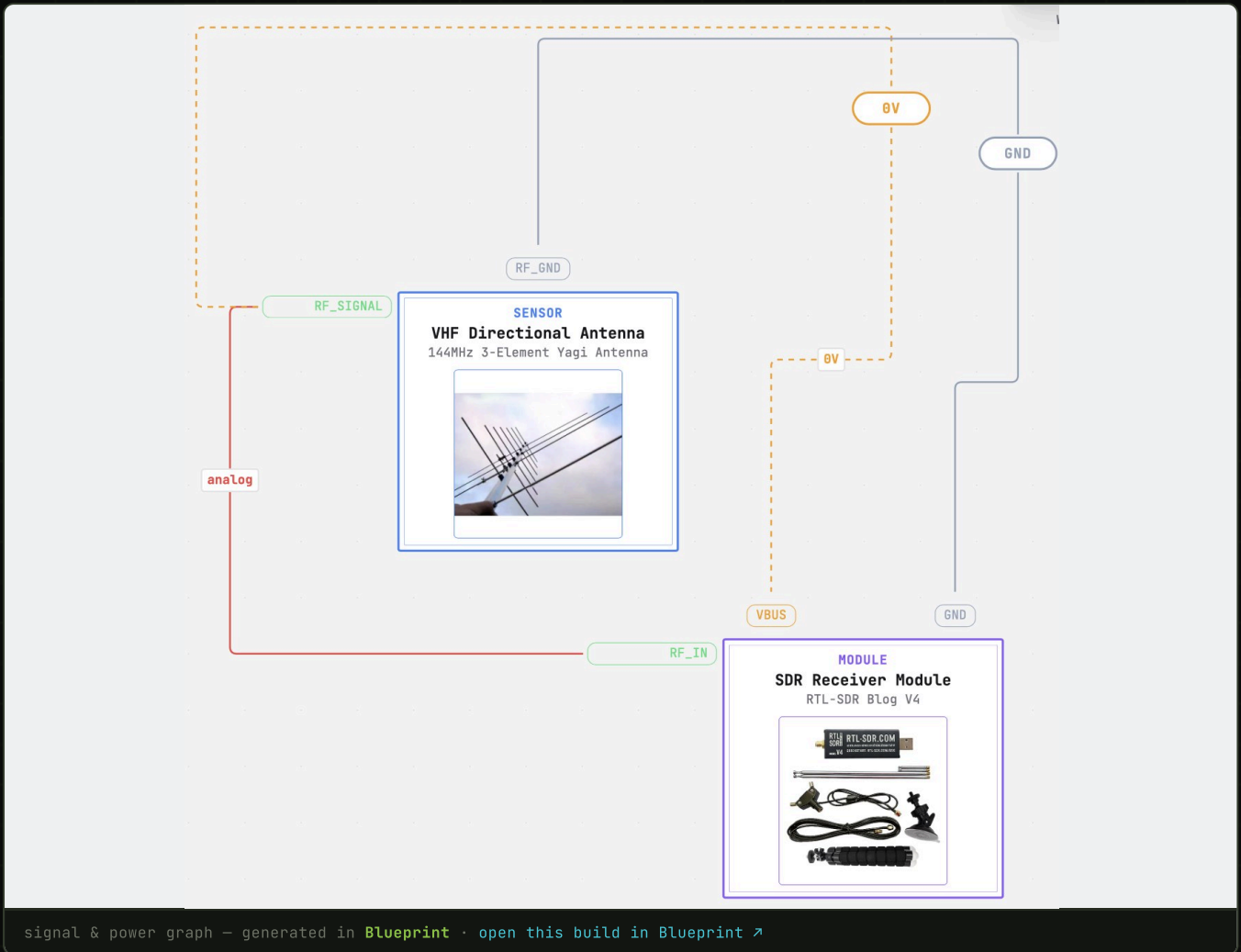
5 Triangulate to a fix

Take bearings from two or three separate spots, draw the lines on a map, and find the intersection. You've located the transmitter — the essence of DF.

interview questions

- What are the main techniques for radio direction finding?
- How does phase-based angle-of-arrival determine a bearing?
- Why must the receivers in a phase-based DF system be coherent?
- Why is multipath the central challenge in real-world DF?
- How do you turn one or more bearings into a transmitter location?
- What are the tradeoffs between RSSI, pseudo-Doppler, and coherent DF?

full wiring schematic



Magnetic Anomaly Detector

Sense a hidden vehicle by the dent it makes in Earth's magnetic field

\$52

intermediate

1 weekend

➔ [open in Blueprint](#)



POWERED BY **Blueprint**

As-built gradiometer — two matched HMC5883L magnetometers on the ends of a non-magnetic boom, ESP32 + OLED at centre.

cost — ~\$52 as built · a two-sensor gradiometer on a non-magnetic boom

why this matters

A large steel object — a vehicle, a pipe, a ship — distorts the Earth's magnetic field around it, and a sensitive magnetometer can detect that distortion from a distance, with no contact and no emission. This is magnetic anomaly detection (MAD), used in defense for submarine hunting and mine detection, and in the civilian world for archaeology, geophysics, and utility locating. Building one teaches you to pull a tiny anomaly out of a large, noisy background field — a purely passive sensing skill with the same signal-in-noise flavour as the RF work, but in the magnetic domain.

WHAT THIS PROVES TO A RECRUITER

You understand magnetometry (fluxgate and magnetoresistive sensors), the Earth's field as a baseline, gradiometry to cancel common-mode drift, calibration, and anomaly detection against a noisy background. Passive magnetic sensing is used across defense, geophysics, and navigation.

➔ the job this maps to: magnetic sensing and geophysical engineering — defense sensor, survey, navigation, and instrumentation teams.

how it works

- **The Earth's field as baseline:** The geomagnetic field is roughly 25–65 microtesla everywhere. A ferrous object perturbs it locally by a tiny amount, so detecting the object means measuring that small change against the large steady baseline.
- **Measuring the field:** A fluxgate or magnetoresistive magnetometer outputs the field along each axis. Cheap 3-axis modules resolve well under a microtesla — enough to see a car-sized anomaly at short range.

- ▶ **Anomaly, not absolute:** What matters is the deviation from the local baseline. You establish the ambient field, then watch for the perturbation as a ferrous mass comes near — the anomaly is the signal.
- ▶ **Gradiometry kills drift:** The baseline itself wanders with temperature, solar activity, and nearby currents. Two matched sensors a fixed distance apart, subtracted, reject that shared drift and respond only to the steep local gradient of a nearby object.
- ▶ **Range and mass:** The anomaly falls off steeply with distance (roughly as the cube), so detection range depends on the object's mass and your sensitivity. Understanding that falloff sets realistic expectations for any magnetic sensor.

BUILD NOTE

A single magnetometer drifts with temperature and Earth-field changes. Use two identical sensors as a gradiometer — subtracting one from the other cancels the common background and leaves the local anomaly, the trick that makes real MAD work.

bill of materials

COMPONENT	PART / SPEC	QTY	~COST
ELECTRICAL			
Main Controller	ESP32 Development Board	1	\$8.00
Front Magnetometer	HMC5883L 3-Axis Magnetometer Module	1	\$5.00
Rear Magnetometer	HMC5883L 3-Axis Magnetometer Module	1	\$5.00
Gradient Display	0.96 inch I2C OLED Display	1	\$4.00
Data Logger Module	MicroSD Card Adapter Module	1	\$3.00
System Power Bank	5V USB Portable Power Bank	1	\$10.00
PRINTED PARTS & HARDWARE			
Sensor Boom	1/2 in. x 2 ft. PVC Sch. 40 Pipe	1	\$3.50
Control Enclosure	3D-printed part	1	\$2.50
Front Sensor Mount	3D-printed part	1	\$0.50
Rear Sensor Mount	3D-printed part	1	\$0.50
M3 Nylon Screws	M3x12mm Nylon Phillips Pan Head Screws	20	\$3.00
M3 Nylon Nuts	M3 Nylon Hex Nuts	20	\$2.00
Velcro Cable Ties	8-inch Reusable Cable Ties	6	\$0.90
Gradient Display Mount	3D-printed part	1	\$2.00
Data Logger Module Mount	3D-printed part	1	\$2.00
Build total (as specified)			\$51.90

♦ BOM, wiring & render generated in **Blueprint** — component links via [blueprint.am](#)

wiring at a glance

SIGNAL & POWER CONNECTIONS

- Front + rear HMC5883L share one I²C bus to the ESP32 (3.3 V + GND)
- 0.96" OLED on the same I²C bus (SDA / SCL)
- microSD logger on SPI for gradient logs
- USB power bank → ESP32 5 V; sensors on the 3.3 V rail
- Sensors fixed apart on a 1/2" PVC boom — nylon hardware only, no ferrous metal

build steps

1 Read the field

Wire a 3-axis magnetometer over I2C and read the field magnitude. Move a steel tool near it and watch the reading jump — your first anomaly.

2 Establish a baseline

Log the ambient field in a clear area and note how much it drifts over minutes. This baseline noise sets the smallest anomaly you can trust.

3 Build a gradiometer

Mount two matched sensors on a non-magnetic boom a fixed distance apart and subtract their readings. Confirm the common drift cancels while a nearby object still shows up strongly.

4 Sweep for objects

Walk the gradiometer over the ground (or past a parked car) and log the gradient. Watch a clear peak appear over buried or nearby ferrous mass.

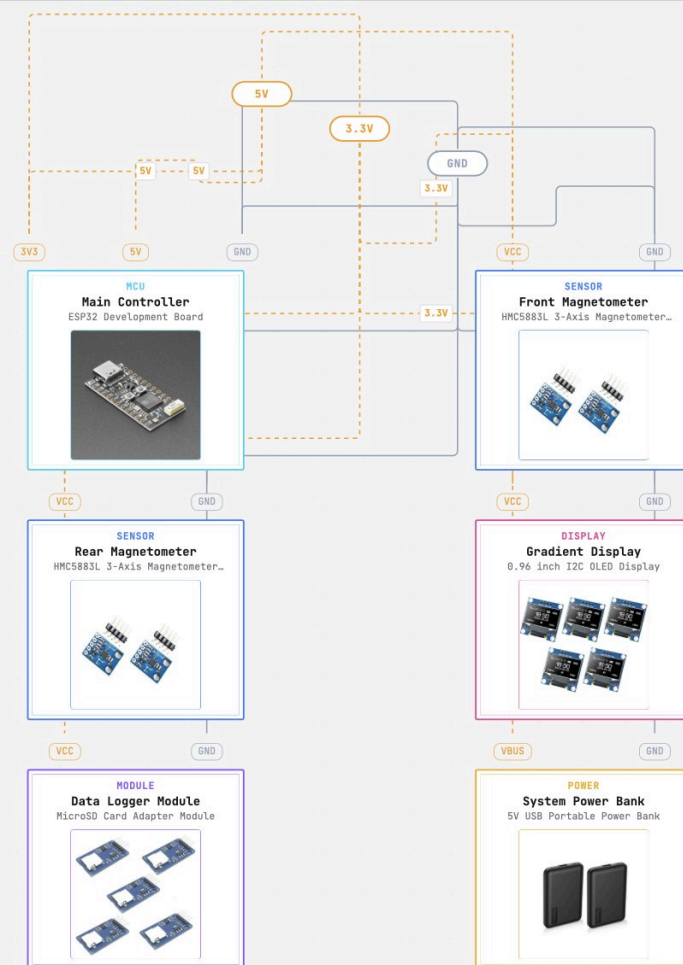
5 Map an anomaly

Take readings on a grid and plot the field or gradient as a heatmap, revealing the shape and location of the hidden mass. That map is the deliverable.

interview questions

- How does a ferrous object create a detectable magnetic anomaly?
- How does detection range scale with object size and distance?
- What is the typical strength of the Earth's field, and why does it matter here?
- What sources of magnetic noise limit a real detector?
- Why is a gradiometer better than a single magnetometer for anomaly detection?
- What is the difference between a fluxgate and a magnetoresistive sensor?

full wiring schematic



signal & power graph – generated in [Blueprint](#) · [open this build in Blueprint](#) ↗

Seismic Unattended Ground Sensor

Feel footsteps and vehicles through the ground before you can see them

\$42

intermediate

1 weekend

➔ [open in Blueprint](#)



POWERED BY **Blueprint**

As-built ground sensor — SM-24 geophone and AD623 amp sealed in an IP65 body, driven into the ground on an M8 spike.

cost — ~\$42 as built • SM-24 geophone + AD623 + ESP32 in a sealed IP65 box

why this matters

Long before you can see or hear an approaching person or vehicle, the ground carries their vibration. A geophone — the same sensor used in oil exploration and earthquake monitoring — turns that ground motion into a voltage, and with the right processing you can detect footsteps, tell a person from a vehicle, and estimate direction. This is the principle behind unattended ground sensors used for perimeter monitoring. It is a passive, emission-free build that teaches low-frequency signal detection, simple classification, and the reality of picking a faint signal out of environmental noise.

WHAT THIS PROVES TO A RECRUITER

You understand seismic and vibration sensing, very-low-frequency signal conditioning, event detection and simple classification (footstep versus vehicle by frequency and cadence), and sensor siting. Passive ground sensing is used in perimeter security, seismology, and structural monitoring.

➔ the job this maps to: **unattended-sensor and perimeter-security engineering** — defense sensor, seismic-monitoring, and structural-health teams.

! YOUR OWN PERIMETER

This is passive monitoring of ground you own or are responsible for — a driveway, a fence line, a site perimeter. Keep it on your own property and respect the privacy of anyone nearby; it senses motion, not identity.

how it works

- **The geophone:** A geophone is a coil-and-magnet transducer that outputs a voltage proportional to ground velocity. Footsteps and vehicles couple energy into the ground as low-frequency waves the geophone picks up.

- **The frequency band:** Seismic events of interest live at low frequencies — footsteps a few to tens of Hz, vehicles lower and more continuous. Amplify and band-pass this range while rejecting wind, mains hum, and higher-frequency noise.
- **Detecting an event:** A footstep is a short impulsive burst; the detector watches signal energy rise above the background floor and flags an event, much like triggering on a noise-adaptive threshold.
- **Classifying the source:** A person produces distinct, spaced impulses at a walking cadence; a vehicle produces continuous, lower-frequency energy. Looking at frequency content and rhythm lets you tell them apart — simple classification from a single sensor.
- **Siting and range:** Coupling to firm ground, sensor depth, and soil type dominate range and sensitivity. Where and how you plant the sensor matters as much as the electronics — the practical heart of unattended sensing.

bill of materials

COMPONENT	PART / SPEC	QTY	~COST
ELECTRICAL			
Main Microcontroller	ESP32-WROOM-32D DevKitC	1	\$8.00
Seismic Sensor	SM-24 4.5Hz Geophone Element	1	\$15.00
High-Gain Amplifier	AD623 Instrumentation Amplifier Module	1	\$5.00
PRINTED PARTS & HARDWARE			
Weatherproof Enclosure	IP65 Waterproof Junction Box	1	\$7.50
Ground Coupling Spike	M8 Stainless Steel Threaded Rod 250mm	1	\$4.50
Spike Locking Nuts	M8 Stainless Steel Hex Nuts	2	\$0.30
Geophone Coupler	3D-printed part	1	\$0.60
MCU Mounting Sled	3D-printed part	1	\$0.25
Amplifier Bracket	3D-printed part	1	\$0.15
M3 Board Screws	M3x8mm Pan Head Screws	8	\$0.40
Build total (as specified)			\$41.70

♦ BOM, wiring & render generated in **Blueprint** — component links via [blueprint.am](#)

wiring at a glance

SIGNAL & POWER CONNECTIONS

- SM-24 geophone (±) → AD623 IN+ / IN-
- AD623 OUT → ESP32 ADC (VP / GPIO36)
- AD623 on 3.3 V; REF tied to GND; gain set by RG
- Band-pass ~2–50 Hz ahead of the ADC (passive RC or active)
- M8 stainless spike couples ground motion into the sealed IP65 box

build steps

- 1 Read the geophone**
Amplify the geophone output and digitize it. Tap the ground nearby and watch the waveform respond — your first ground-motion signal.
- 2 Condition the band**
Band-pass to roughly 2–50 Hz to keep footstep and vehicle energy while rejecting wind and mains. Confirm a quiet baseline when nothing is moving.
- 3 Detect footsteps**
Compute short-window energy and flag events above a noise-adaptive threshold. Walk toward the sensor and count how far away it first triggers.
- 4 Classify person vs vehicle**
Compare the frequency content and rhythm of footsteps versus a rolling vehicle, and build a simple rule that labels each event type.

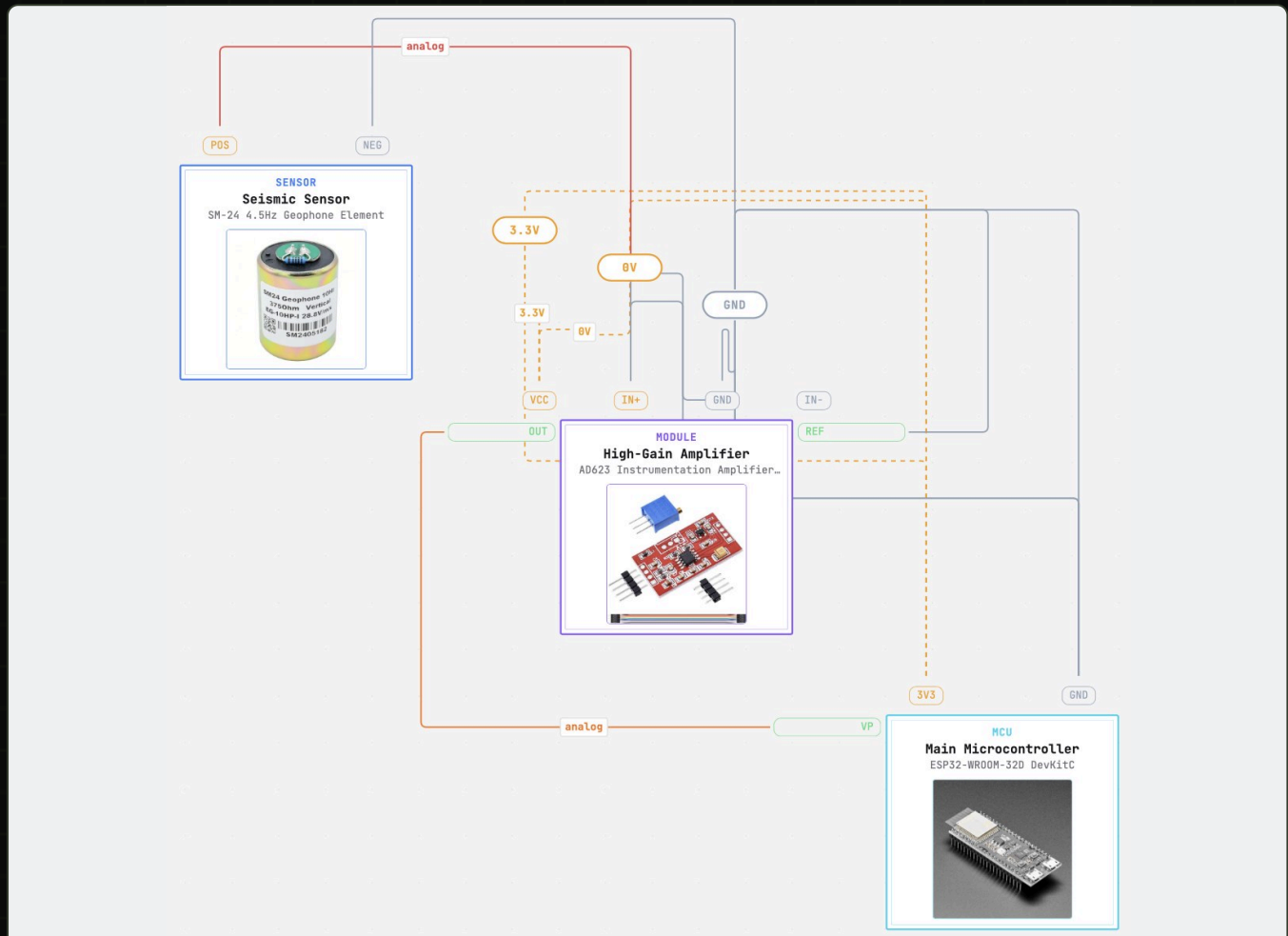
5 Test siting and range

Move the sensor between soft and firm ground and re-measure detection range. Document how coupling changes everything — the real lesson of ground sensing.

interview questions

- How does a geophone convert ground motion into a signal?
- What frequency band matters for detecting footsteps and vehicles?
- How would you detect a discrete seismic event against background noise?
- How can you distinguish a person from a vehicle with one sensor?
- Why does sensor siting dominate detection range?
- What environmental sources limit a seismic sensor's sensitivity?

full wiring schematic



signal & power graph – generated in [Blueprint](#) · [open this build in Blueprint](#)

Drone Detection Receiver

Spot a drone by the radio it can't stop transmitting — passive counter-UAS

\$500

advanced

1-2 weekends

➔ open in Blueprint



POWERED BY **Blueprint**

As-built counter-UAS receiver — HackRF One + Raspberry Pi 4, motorised pan/tilt directional antenna, status LCD. Receive-only.

cost — ~\$500 as built (HackRF One + Pi 4 + pan/tilt) • from ~\$40 with an RTL-SDR for 2.4 GHz only

why this matters

Small drones have gone from novelty to a serious security concern, and detecting them is now a major defense and infrastructure problem. Most drones constantly transmit — a control link on 2.4 or 5.8 GHz and usually a video downlink — and those emissions are a fingerprint you can detect passively with an SDR. Building a drone-detection receiver teaches you to scan the ISM bands, recognise the signatures of drone control and video protocols, and flag a drone's presence without transmitting anything yourself. It is a receive-only, topical build that sits right at the centre of the fast-growing counter-UAS field.

WHAT THIS PROVES TO A RECRUITER

You understand wideband spectrum scanning, recognising signal signatures (frequency-hopping control links, video downlinks), the ISM bands drones use, and energy and feature detection — the passive-sensing core of counter-UAS. It maps directly onto RF-sensing and spectrum-awareness roles.

➔ the job this maps to: counter-UAS and RF-sensing engineering — drone-defense, critical-infrastructure-protection, and spectrum-monitoring teams.

! DETECT ONLY — NEVER JAM OR HIJACK

This build never transmits. Jamming, hijacking, or spoofing a drone or its GPS is illegal in most countries and can be dangerous, so this stays entirely passive — you sense that a drone is present, nothing more. Respect privacy and spectrum law.

how it works

- **Drones are noisy on purpose:** A drone needs a control link and usually a live video feed, so it is almost always transmitting on 2.4 GHz, 5.8 GHz, or both. Those emissions are exactly what a passive receiver looks for.
- **Scanning the bands:** An SDR sweeps the ISM bands and builds a spectrogram, or waterfall. Drone control links often frequency-hop across the band in a recognisable pattern, and video downlinks appear as wide, steady carriers.
- **Signatures, not just energy:** Distinguishing a drone from WiFi and Bluetooth means recognising the shape and timing of its signals — the hopping cadence of the control link, the bandwidth of the video — rather than just seeing energy in the band.
- **Detection and alerting:** When a matching signature appears, you flag it. Advanced systems classify the drone type from its protocol fingerprint; a first build can simply and reliably say 'a drone-like signal is present.'
- **Range and antennas:** A directional antenna extends range and, swept, hints at direction. Combined with the direction finder from project 2, passive detection can become passive location — all without emitting a thing.

bill of materials

COMPONENT	PART / SPEC	QTY	~COST
ELECTRICAL			
Main Processor	Raspberry Pi 4 Model B 4GB	1	\$55.00
Wideband SDR	HackRF One	1	\$340.00
Pan Servo Motor	MG996R Servo	1	\$12.00
Tilt Servo Motor	MG996R Servo	1	\$12.00
Servo Controller	PCA9685 16-Channel PWM Module	1	\$7.00
Battery Power Supply	Geekworm UPS HAT	1	\$35.00
System Status Display	3.5 inch SPI TFT LCD	1	\$22.00
PRINTED PARTS & HARDWARE			
Main Enclosure Base	3D-printed part	1	\$3.50
Main Enclosure Lid	3D-printed part	1	\$1.50
Raspberry Pi Mount	3D-printed part	1	\$0.50
HackRF Mount	3D-printed part	1	\$0.75
Status LCD Bezel	3D-printed part	1	\$0.30
Pan Servo Bracket	3D-printed part	1	\$0.60
Tilt Servo U-Bracket	3D-printed part	1	\$0.80
Directional Antenna Platform	3D-printed part	1	\$1.20
Pan Thrust Bearing	Uxcell F623ZZ Flange Bearing (or similar 40mm Thrust Bearing)	1	\$4.50
M3 Machine Screws	M3x12mm Socket Head Cap Screws	24	\$1.20
M3 Heat-Set Inserts	M3x5x4mm Brass Heat Set Inserts	24	\$1.92
Servo Controller Mount	3D-printed part	1	\$2.00
Build total (as specified)			\$501.77

♦ BOM, wiring & render generated in **Blueprint** — component links via [blueprint.am](#)

wiring at a glance

SIGNAL & POWER CONNECTIONS

- HackRF One (wideband, 2.4 + 5.8 GHz) → Raspberry Pi 4 over USB
- PCA9685 PWM driver on I²C; drives pan + tilt M6996R servos (5 V from UPS)
- 3.5" SPI TFT status display (MOSI / SCLK) on the Pi
- Geekworm UPS HAT powers Pi + SDR (5 V); telemetry on I²C
- Directional (log-periodic / panel) antenna on the tilt platform – receive only

build steps

1 Watch the band

Point an SDR at 2.4 GHz and open a waterfall display. Identify the ambient WiFi and Bluetooth so you know the normal background.

2 Fingerprint a drone

With a drone you own powered on nearby (legally), find its control link and video downlink in the waterfall. Note the hopping pattern and the video carrier's width.

3 Automate detection

Write logic that watches for that signature — the hopping cadence or the video carrier — and raises a flag when it appears, ignoring ordinary WiFi.

4 Extend to 5.8 GHz

Add a 5.8 GHz-capable front end (or a HackRF) to catch drones that use the upper band, and confirm you detect both control and video.

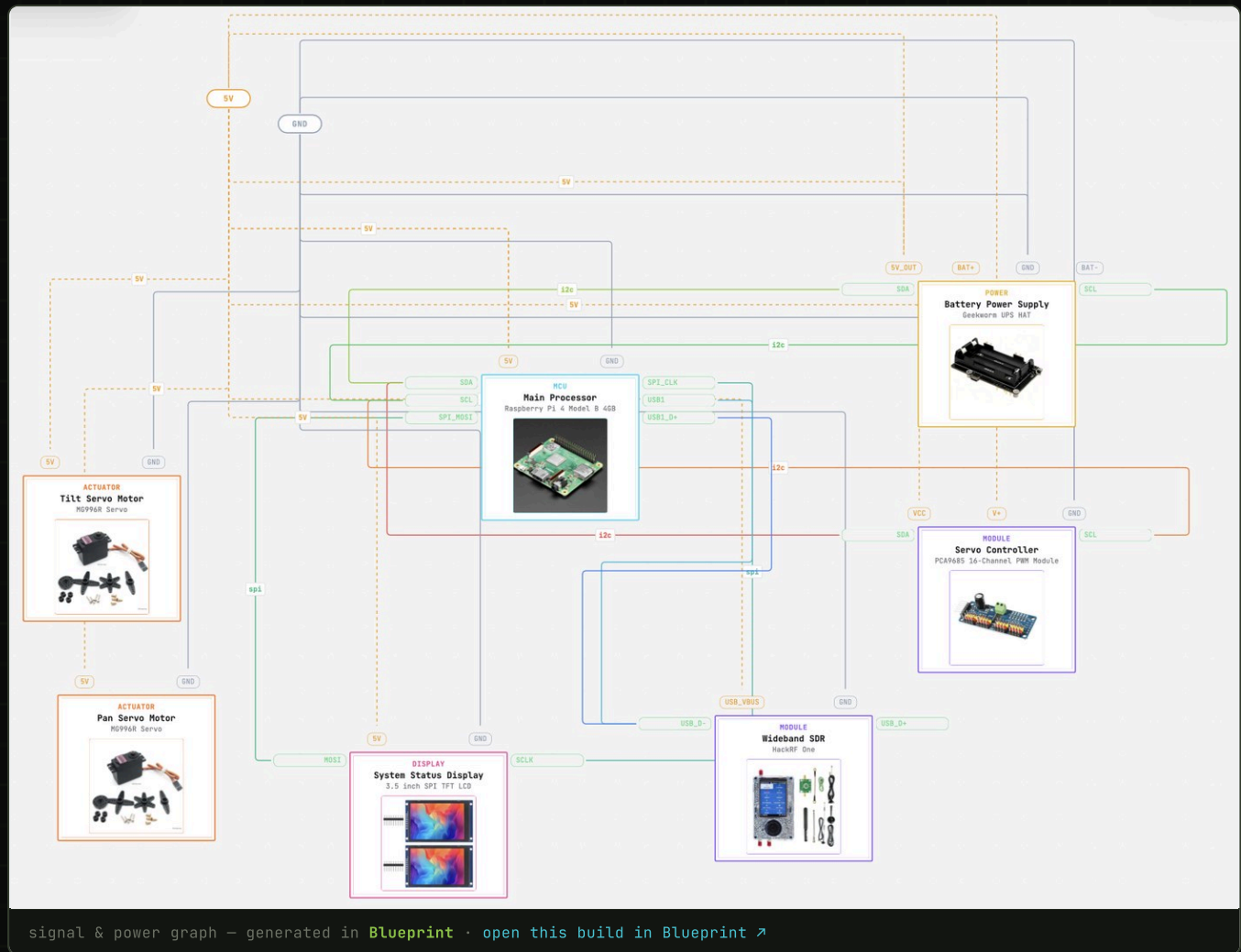
5 Add a directional antenna

Swap to a directional antenna to extend range and get a rough bearing, and note how detection distance improves. Detection stays entirely passive.

interview questions

- Why are most drones detectable by their RF emissions?
- What is a waterfall / spectrogram, and why is it central to this build?
- What bands do drone control and video links typically use?
- Why is passive detection preferable (and legal) versus active countermeasures?
- How do you distinguish a drone signal from WiFi or Bluetooth?
- How could passive detection be extended to locate a drone?

```
## full wiring schematic
```



Frequency-Hopping Encrypted Link

A radio link that scatters across the band and encrypts every packet — resilient comms by design

\$46

advanced

1-2 weekends

➤ open in Blueprint



POWERED BY **Blueprint**

As-built pair — two identical ESP32 + nRF24L01+ nodes with OLED and Next/Send keys; AES + hopping in firmware.

cost — ~\$46 as built • a matched pair of handheld nodes (Node A + Node B)

why this matters

In contested environments, communications have to survive both interception and interference, and the classic answers are spread spectrum and encryption. Build a data link that hops across many frequencies in a pattern only the two endpoints know, and encrypts every packet with AES, and you have a channel that is hard to jam, hard to intercept, and hard to even detect — the same principles behind military tactical radios (and, historically, a patent from Hedy Lamarr). It is a superb lesson in resilient communications: synchronisation, spread spectrum, and applied cryptography, all on cheap radio modules.

WHAT THIS PROVES TO A RECRUITER

You understand frequency-hopping spread spectrum, transmitter and receiver hop synchronisation, why spreading resists jamming and interception, and applying authenticated encryption (AES) to a real link. Resilient, secure communications is a core defense and aerospace EE discipline.

➔ the job this maps to: secure and resilient communications engineering — tactical-radio, aerospace-datalink, and secure-IoT teams.

! SECURE YOUR OWN LINK, AND STAY LEGAL

Build this to protect communications you are allowed to operate, and stay within your local radio regulations — the legal band, power limits, and any licensing. Resilient comms is about protecting your own link, not evading lawful authority. And never roll your own cryptography for anything that matters; use a vetted AES implementation.

how it works

- **Spread spectrum, why:** Concentrating a signal on one frequency makes it easy to jam and easy to intercept. Spreading it — hopping across many channels — means a jammer must cover the whole band and an eavesdropper must know where to listen, dramatically improving resilience.
- **Frequency hopping:** Both ends step through a sequence of frequencies many times a second, following the same pseudo-random pattern seeded by a shared secret. To anyone without the pattern, the signal appears as brief, scattered blips.
- **Synchronisation is everything:** The receiver must hop in lockstep with the transmitter. A shared clock or a sync handshake keeps them aligned; lose sync and the link drops — which is why timing is the hardest part of the build.
- **Encrypting the payload:** Hopping hides where you transmit; encryption hides what you send. Wrapping each packet in authenticated AES means even a receiver that follows the hops cannot read or forge the contents without the key.
- **What the pros add:** Real tactical radios add much faster hopping, wider bands, error correction, and rigorous key management. Your build captures the core ideas honestly; the rest is scale and rigour.

bill of materials

COMPONENT	PART / SPEC	QTY	~COST
ELECTRICAL			
Node A Microcontroller	ESP32 DevKit V1	1	\$6.00
Node A Transceiver	nRF24L01+ PA LNA Wireless Module	1	\$4.50
Node A Interface Display	0.96 inch I2C OLED Display	1	\$3.50
Node A Next Button	6x6mm Tactile Push Button	1	\$0.10
Node A Send Button	6x6mm Tactile Push Button	1	\$0.10
Node B Microcontroller	ESP32 DevKit V1	1	\$6.00
Node B Transceiver	nRF24L01+ PA LNA Wireless Module	1	\$4.50
Node B Interface Display	0.96 inch I2C OLED Display	1	\$3.50
Node B Next Button	6x6mm Tactile Push Button	1	\$0.10
Node B Send Button	6x6mm Tactile Push Button	1	\$0.10
PRINTED PARTS & HARDWARE			
Node A Enclosure Shell	3D-printed part	1	\$1.50
Node B Enclosure Shell	3D-printed part	1	\$1.50
Node A Faceplate	3D-printed part	1	\$0.50
Node B Faceplate	3D-printed part	1	\$0.50
Enclosure Assembly Screws	M3x8mm Button Head Hex Screws	16	\$0.80
Threaded Inserts	M3x5x4mm Brass Heat Set Inserts	16	\$1.28
Node A Transceiver Mount	3D-printed part	1	\$2.00
Node A Next Button Mount	3D-printed part	1	\$2.00
Node A Send Button Mount	3D-printed part	1	\$2.00
Node B Transceiver Mount	3D-printed part	1	\$2.00
Node B Next Button Mount	3D-printed part	1	\$2.00
Node B Send Button Mount	3D-printed part	1	\$2.00
Build total (as specified)			\$46.48

wiring at a glance (per node)

SIGNAL & POWER CONNECTIONS

- ESP32 ↔ nRF24L01+ (PA/LNA) over SPI — SCK D18, MOSI D23, MISO D19, CSN D5, CE D4
- 0.96" OLED on I²C — SDA D21, SCL D22
- Next / Send buttons on D32 / D33 to GND
- Radio + display on 3.3 V; Node A and Node B are identical
- AES + hop schedule seeded by a shared key in firmware

build steps

1 Build a plain link

Get two modules talking on a single fixed frequency first — send and receive packets reliably. Prove the basic radio before adding complexity.

2 Add channel hopping

Have both ends step through a list of channels on a shared schedule. Start slow, confirm they stay in sync, and watch packets get through as they hop.

3 Synchronise robustly

Add a sync handshake so a receiver can find and lock onto the transmitter's hop sequence, and recover if it drifts. This is the make-or-break step.

4 Seed the pattern with a secret

Generate the hop sequence from a shared key so only the paired endpoints know the pattern. Confirm an unpaired receiver sees only scattered blips.

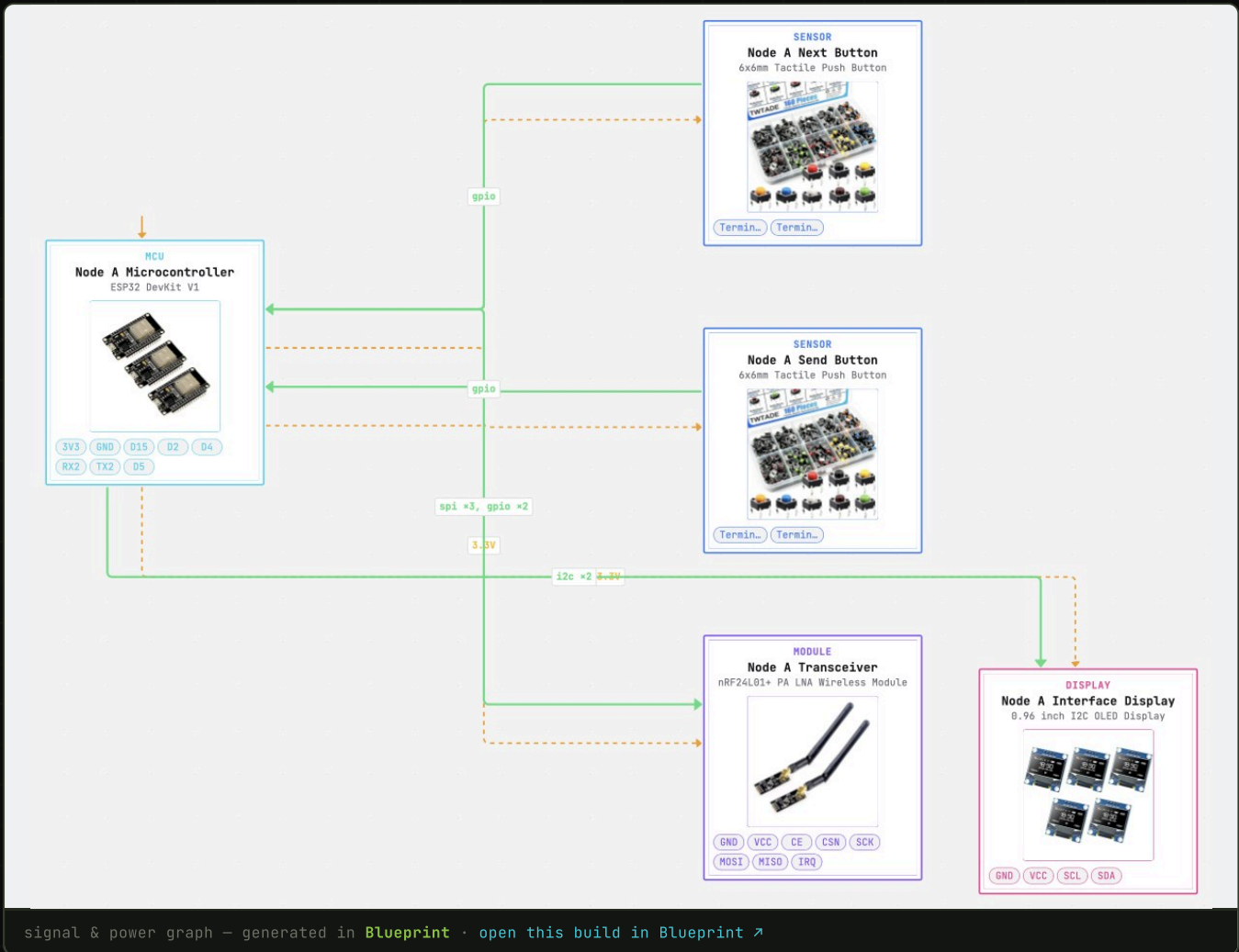
5 Encrypt every packet

Wrap each payload in authenticated AES with the shared key. Verify the link still works end to end and that captured packets are unreadable without the key.

interview questions

- Why does spread spectrum resist both jamming and interception?
- How does frequency-hopping spread spectrum work?
- Why is synchronisation the central challenge in a hopping link?
- What does encryption add on top of frequency hopping?
- Why should you never implement your own cryptographic primitives?
- What do real tactical radios add beyond a hobby hopping link?

full wiring schematic



› wrap

Defense EE isn't about weapons

— it's about who sees first.

A receiver pulling position from space, a needle pointing at a hidden transmitter, a dent in the magnetic field, footsteps through the ground, a drone's own radio giving it away, a link that scatters and encrypts itself — six ways to detect, locate, navigate, and protect, without emitting harm. Every one is receive-only, passive, or protective; the value is in the signal processing: the correlation, the phase, the anomaly detection, the synchronisation. That is exactly what defense, aerospace, and spectrum teams hire for.

the skill ladder

Build them in order and each one hands you the technique the next one needs:

SKILL LADDER

GNSS → direction finding → magnetic anomaly → seismic → passive RF detection → secure comms

! THE LINE THIS GUIDE STAYS ON

Everything here is receive-only, passive, or protective — built to detect, locate, navigate, and secure, never to jam, spoof, target, or harm. Jamming or spoofing GNSS or drones is illegal and dangerous. Follow your local radio and privacy laws, keep the one transmitter here (the hopping link) legal and within its band, and don't point these tools at people. Build to understand, and to protect.

DO THIS

Build even one. Capture it — a position track, a bearing swinging to a hidden transmitter, a magnetic anomaly map, a footstep trigger, a drone flagged on a waterfall — and put the code on GitHub with your method, your results, and an honest note on scope and legal limits. One sensing demo and one clean repo prove you can do defense-grade signal work.

Full build notes, parts links, and the next field guide go out in The Build List. The people actually shipping these — and trading antenna and coherence tricks — are in Ohm's Club.

◆ BUILT WITH BLUEPRINT

Every schematic, bill of materials and 3D render in this field guide came out of **Blueprint** by describing each build in plain English — no manual part-hunting or schematic work. Design your own unit at blueprint.am.

— Enes (@1nska) · Electrical Engineer